

Bhawani Lal

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leetcode.com/bhawani01 | bhawani-portfolio.vercel.app

EDUCATION

University of Strathclyde

Glasgow, UK

MSc Advanced Computer Science with Data Science

Jan 2026 – Jan 2027

- Dissertation: SmartMaintain — Predictive Maintenance System using LSTM, Random Forest & SHAP on NASA C-MAPSS dataset
- Relevant modules: Machine Learning, Data Analytics, Distributed Information Systems, Advanced Algorithms

IIMT University

Meerut, India

Bachelor of Science in Computer Science — GPA: 7.26 / 10.0

Aug 2022 – Sep 2025

EXPERIENCE

Crowd Operations and Safety Specialist

Jun 2026 – Present

G4S Events Scotland

Glasgow, UK

- Coordinated crowd control operations and threat-mitigation protocols across high-footfall Glasgow venues, managing safety flow for events with over 10,000+ attendees.
- Applied ACT (Action Counter Terrorism) frameworks to systematically identify, assess, and report security risks, ensuring 100 percent compliance with UK safety regulations.
- Liaised directly with emergency services and senior event management to execute real-time incident response plans under high-pressure scenarios.

PROJECTS

SmartMaintain – Predictive Maintenance System | *Python, LSTM, Random Forest, SHAP, RAG* July 2026 - Present

- MSc dissertation: built a hybrid LSTM + Random Forest pipeline on NASA C-MAPSS FD001 dataset to predict Remaining Useful Life (RUL) of turbofan engines, achieving MAE of 12.76 and RMSE of 16.21
- Engineered 51 features via rolling statistics and lag windows; applied RUL capping at 125 cycles for realistic training targets
- Conducted SHAP explainability analysis identifying top predictive sensors (mean_s3, mean_s21, last_s11)
- Developing a SmartMaintain Copilot using RAG and function calling to enable natural language querying of maintenance insights over trained model outputs

AI Document Intelligence Tool | *Python, Claude API, LangChain, RAG, FastAPI*

July 2026 - Present

- Built a RAG-powered tool that ingests business documents (PDFs, reports) and enables natural language Q&A over their content
- Implemented semantic chunking, vector embeddings, and retrieval-augmented generation using the Claude API
- Exposed a FastAPI backend with a React frontend for document upload, query, and structured insight extraction
- Targeting use-cases aligned with McKinsey Digital's AI-driven document analysis and knowledge management work

Financial Risk Analytics Dashboard | *Python, Streamlit, Plotly, Pandas, yfinance*

July 2026 – Present

- Built a real-time financial risk dashboard pulling live equity data via yfinance API, covering 10 stocks with dynamic filtering by ticker and date range
- Implemented rolling volatility analysis (30-day window), asset correlation heatmap, and Value at Risk (VaR) calculation at 95% confidence interval
- Engineered end-to-end data pipeline with Pandas for cleaning and aggregation, Plotly for interactive visualisation — deployed live on Streamlit Cloud [**Live Demo**]
- Designed for financial analysts to surface risk exposure and portfolio correlation insights — directly relevant to quantitative analytics in financial services

3D Cyberpunk Portfolio Website | *React, Three.js, React Three Fiber, Vite*

June 2026

- Designed and deployed a high-performance interactive 3D web application utilizing React, Three.js, and React Three Fiber to showcase technical projects and data visualizations.
- Optimized asset rendering pipelines and CSS delivery configurations, minimizing load times and ensuring seamless cross-browser compatibility across desktop and mobile platforms.
- Deployed on Vercel with continuous deployment via GitHub — [live link](#)

CERTIFICATIONS

JPMorgan Chase Advanced Software Engineering Job Simulation <i>Forage</i> Credential ID: NWqBfQBwhB7GpA28z	<i>May 2026</i>
HPE Software Engineering Job Simulation <i>Forage</i>	<i>Jun 2026</i>
Building with the Claude API <i>Anthropic Academy</i>	<i>July 2026</i>
Claude Platform 101 <i>Anthropic Academy</i>	<i>June 2026</i>
AI Fluency: Framework & Foundations <i>Anthropic Academy</i>	<i>June 2026</i>
Python for Data Science, AI and Development <i>IBM – Coursera</i>	<i>Jan 2025</i>
Introduction to Software Engineering <i>IBM – Coursera</i>	<i>Dec 2024</i>

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, SQL, R, C/C++, HTML/CSS
Frameworks: Spring Boot, React, Flask, Tailwind CSS, FastAPI, LangChain
AI / ML: LSTM, Random Forest, SHAP, RAG, CNN, NLP, Claude API, TensorFlow, OpenCV
Technologies: Apache Kafka, REST APIs, Git, Maven, Docker, JPA/Hibernate, Three.js
Databases: MySQL, MongoDB, PostgreSQL, H2
Tools: IntelliJ IDEA, VS Code, GitHub, Overleaf, Streamlit, Vercel